

VELOCITY OF ANEALING PATHS

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$$\dot{\pi}_{\beta}(x) \coloneqq \frac{\mathrm{d}}{\mathrm{d}\beta} \log \pi_{\beta}(x)$$

► We will abuse notation and will identify π_β with both the density or distributional derivative

► In summary, we can equivalently express the velocity $\dot{x}_\beta$ terms of densities and expectation

$$\dot{\pi}_{\beta}[f] \coloneqq \frac{\mathrm{d}}{\mathrm{d}\beta} \pi_{\beta}[f]$$

► **Velocity as a function:** $x \mapsto \dot{\pi}_\beta(x)$ defines a function equals to the instantaneous percentage change in density $\pi_\beta(x)$ at β for all $x \in X$

► **Velocity as a measure:** $f \mapsto \dot{\pi}_\beta[f]$ defines a (signed)-measure equal to the instantaneous rate of change in expectation $\pi_\beta[f]$ at β for $f : \mathbb{X} \mapsto \mathbb{R}$ with $f(x)\dot{\pi}_\beta(x)$ integrable with respect to π_β :

$$\dot{\pi}_{\beta}[f] = \pi_{\beta}\left[f\dot{\pi}_{\beta}\right]$$

▶ We can relate the derivative of the expectation and densities through the identity:

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- ▶ **Velocity as a function:** $x \mapsto \dot{\pi}_\beta(x)$ defines a function equals to the instantaneous percentage change in density $\pi_\beta(x)$ at β for all $x \in \mathbb{X}$

$$\dot{\pi}_\beta(x) := \frac{d}{d\beta} \log \pi_\beta(x)$$

- ▶ **Velocity as a measure:** $f \mapsto \dot{\pi}_\beta[f]$ defines a (signed)-measure equal to the instantaneous rate of change in expectation $\pi_\beta[f]$ a β for $f : \mathbb{X} \mapsto \mathbb{R}$ with $f(x)\dot{\pi}_\beta(x)$ is integrable with respect to π_β :

$$\dot{\pi}_\beta[f] := \frac{d}{d\beta} \pi_\beta[f]$$

- ▶ We will abuse notation and will identify π_β with both the density or distributional derivative
- ▶ We can relate the derivatives of the expectation and densities through the identity:

$$\dot{\pi}_\beta[f] = \pi_\beta \left[f \dot{\pi}_\beta \right]$$

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